MARYAM REHAN INTERNSHIP CYBER SECURITY (SOC)

WEEK 2 – TASK 4

Linux Basics and Log File Analysis (/var/log)

1. Introduction

Linux is widely used in cybersecurity and Security Operations Centers (SOCs). Log files help analysts monitor system activity, investigate authentication failures, and identify suspicious events.

2. Objective

- Practice basic Linux commands.
- Understand the `/var/log` directory.
- Search logs using `grep`.
- Learn to monitor live logs using `tail -f`.

3. Basic Linux Commands

Commands such as `ls`, `cd`, `pwd`, `cat`, `less`, `tail`, and `grep` help users navigate directories, view files, and search for specific information.

4. Linux Log Files

The `/var/log` directory stores system and service logs. Files such as `auth.log` and `syslog` may contain authentication events and general system messages, depending on the Linux configuration.

5. Log Searching

The following command searches for failed authentication messages:

grep 'Failed' /var/log/auth.log

The `tail` command displays recent log entries, while `tail -f` monitors new entries in real time. Actual results depend on the available log files and recorded events.

6. Security Importance

Log analysis helps SOC analysts investigate repeated login failures, recognize unusual activity, and support incident investigations. Suspicious events should be verified before drawing conclusions.

7. Conclusion

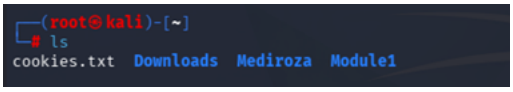
This task introduced essential Linux commands, log file locations, text searching, and live log monitoring. These skills provide a foundation for Linux administration and SOC security monitoring.

Evidence

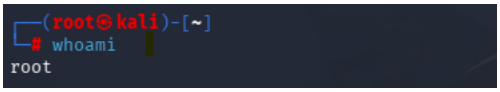
E1 : Demo of 10 Linux commands:



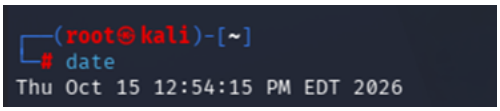
Displays the current working directory



Current directory files folders Listing



Current logged-in username show



System date time show



Demo folder create

```
┌──(root㉿kali)-[~]
└─# cd demo

┌──(root㉿kali)-[~/demo]
└─#
```

Inside Demo folder

```
┌──(root㉿kali)-[~/demo]
└─# touch file.txt
```

Empty text file create

```
┌──(root㉿kali)-[~/demo]
└─# echo "Hello Linux" > file.txt
```

Write text inside file

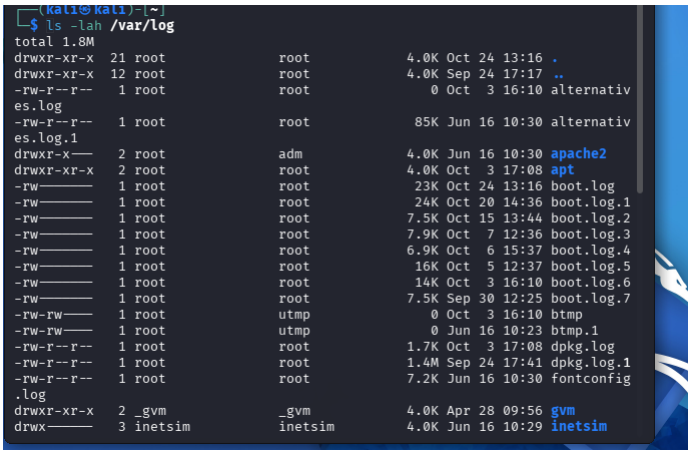
```
┌──(root㉿kali)-[~/demo]
└─# cat file.txt
Hello Linux
```

Display Text inside file

```
┌──(root㉿kali)-[~/demo]
└─# cp file.txt copy.txt
```

Create Copy of file

E2: `/var/log` Directory

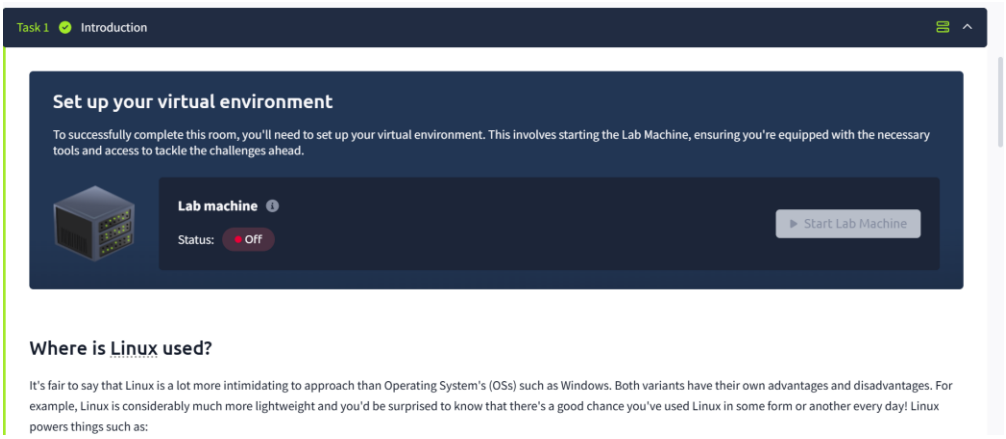


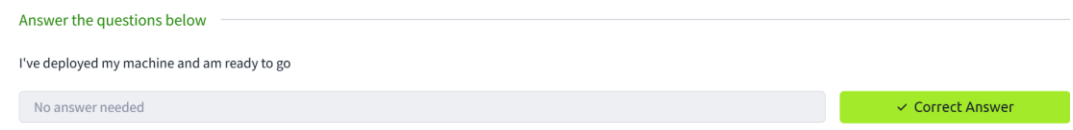
```
.log
-rw-r--r--    1 root             root              61 Oct 15 12:47 macchanger
.log.1.gz
-rw-r--r--    1 root             root              66 Oct  3 15:12 macchanger
.log.2.gz
drwxr-----    2 mosquitto        mosquitto       4.0K May 14 05:43 mosquitto
drwxr-s---    2 mysql            adm             4.0K Jun 16 10:29 mysql
drwxr-xr-x    2 root             adm             4.0K Jun 16 10:28 nginx
drwxr-xr-x    2 _gvm             _gvm            4.0K Apr 16  2026 notus-scan
ner
drwxr-xr-x    2 root             root            4.0K Apr 27 17:25 openvpn
drwxrwxr-t    2 root             postgres        4.0K Jun 16 10:29 postgresql
drwx------    2 root             root            4.0K Jun 16 10:23 private
drwxr-s---    2 redis            adm             4.0K Jun 16 10:29 redis
drwxr-xr-x    4 root             root            4.0K Jun 16 10:25 runit
drwxr-x---    2 root             adm             4.0K May 26 08:46 samba
drwx------    2 speech-dispatcher root            4.0K Jun 16 10:25 speech-dis
patcher
drwxr-xr-x    2 stunnel          stunnel         4.0K Jun 16 10:28 stunnel
drwxr-xr-x    2 root             root            4.0K Jan 15  2024 sysstat
-rw-rw-r--    1 root             utmp             19K Oct 24 13:16 wtmp
-rw-r--r--    1 root             root            8.0K Oct 24 13:16 wtmp.db
-rw-r--r--    1 root             root             36K Oct 24 13:16 Xorg.0.log
-rw-r--r--    1 root             root             36K Oct 24 12:39 Xorg.0.log
.old
```

E3:Failed Login Search &Live Log Monitoring

```
labex:project/ $ grep -iE 'failed|failure|invalid user' /var/log/auth.log
grep: /var/log/auth.log: No such file or directory
labex:project/ $ grep -iE 'failed|failure|error' /var/log/bootstrap.log | tail -n 20
2024-06-27 14:20:16 URL:http://ftpmaster.internal/ubuntu/pool/main/libg/libgpg-error/libgpg-error0_1.43-3_amd64.deb [69684/69684] -> "/build/chroot//var/cac
he/apt/archives/partial/libgpg-error0_1.43-3_amd64.deb" [1]
Selecting previously unselected package libgpg-error0:amd64.
Preparing to unpack .../libgpg-error0_1.43-3_amd64.deb ...
Unpacking libgpg-error0:amd64 (1.43-3) ...
Setting up libgpg-error0:amd64 (1.43-3) ...
labex:project/ $ tail -f /var/log/bootstrap.log
Setting up gpgv (2.2.27-3ubuntu2) ...
Setting up libxxhash0:amd64 (0.8.1-1) ...
Setting up libtasn1-6:amd64 (4.18.0-4build1) ...
Setting up libstdc++6:amd64 (12-20220319-1ubuntu1) ...
Setting up libhogweed6:amd64 (3.7.3-1build2) ...
Setting up libapt-pkg6.0:amd64 (2.4.5) ...
Setting up libp11-kit0:amd64 (0.24.0-6build1) ...
```

TRY HACK ME





Task 2 Talking to Linux

Who are you on this machine?

Linux is a popular choice for servers and the machines that you'll interact within cyber security. This interaction starts with the terminal, not a mouse. In this room, you will interact with your first ever Linux terminal - a place where we can run commands on the system.



Being familiar with the terminal (command line interface) on Linux is a critical skill as you will most likely spend most of your time here in Cyber security. From running hacking tools to hunting down attackers.

A command is an instruction that we can give the computer to perform a given task. One of the first commands we can do is `whoami`. This is important in Cyber security because you will often be changing users on the machine, which determines what you can and cannot do.

`whoami`	tells you who you are on the system

When you run your first command, you will see some text. We call this output. Interacting with Linux feels like a conversation. You give it an instruction, and it gives you output. We can ask Linux to output specific text for us.

What command would we use to find out who we are on the system?

- (●) whoami
- ( ) echo

✓ Correct Answer

What command would we use to output some text?

- ( ) whoami
- (●) echo

✓ Correct Answer

Task 3 Finding Your Way Around

Move around without ever touching a mouse

Let's learn how to navigate around the system's files via the terminal. Four commands do almost all navigation

`ls`	list what's in the current folder
`cd`	change directory — move into a folder
`cat`	show the contents of a file
`pwd`	print working directory — "where am I?"

Run `ls` in the current folder. How many folders are there?

4 ✓ Correct Answer

One of those folders contains a file. Which folder is it?

folder4 Check

Uh-oh! The answer you provided may not be in English. Please review it and try again.

Use `cat` to read the file within this folder. What does it say?

HelloWorld! Check

You are going too fast. Please try again in a little while.

Task 4 Let the Machine Do the Searching

Never search for things by hand again

We can use a set of Linux commands to help us efficiently search for and through files. Rather than scrolling through lots of text, we can have Linux do the hard work

`find`	search for files by their name. For example, `find -name passwords.txt`
`grep`	searches *inside* for text. For example, `grep "password123" passwords.txt`

There's a log file from a web server in your home folder called **access.log,** which is hundreds of lines long. Somewhere inside is a flag.



▶ *Why you're doing this*

Answer the questions below

After using `grep THM access.log` a flag was found. What is the flag?

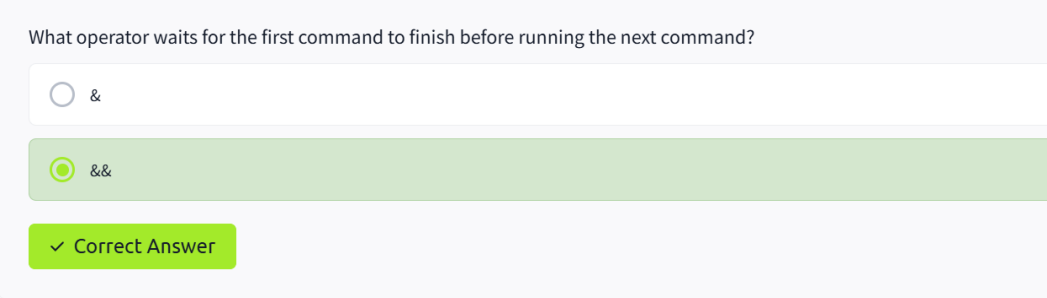
THM{ACCESS} ✓ Correct Answer

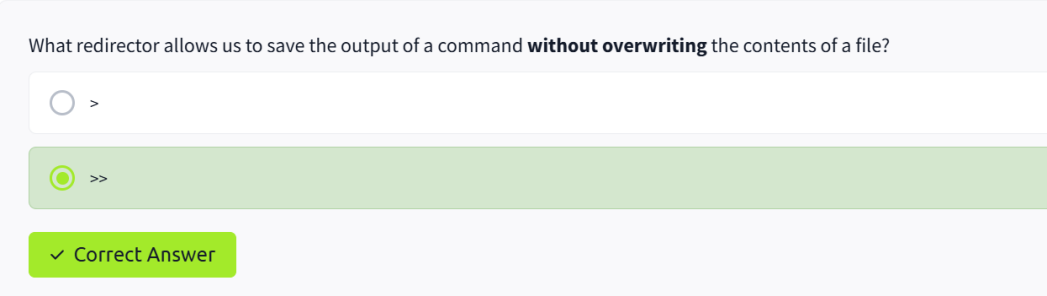
Task 5 Shell Operators (Combining Commands)

Combine commands and capture their output

In Linux, there are a set of special characters that can combine commands together. These are called "Operators" which tell Linux how i being able to combine commands to doing what's called a redirection - sending the output of commands elsewhere. Let's go over these

`&`	Runs the command, but does not wait for it to finish before you can do anything else. The command runs in the backgorund, and is helpful for commands that might take a while to complete, or ones that you want to keep running.
`&&`	Runs both commands, but waits for the first command to finish first, before the next. Like a set of dominoes.







Task Descriptions and Conclusion

TryHackMe – Linux Fundamentals (Pt1)

Task 1: Introduction

This task introduced the Linux Fundamentals learning environment and explained the importance of Linux in information technology and cybersecurity. Linux is widely used in servers, cloud infrastructure, network administration, and security tools. Understanding Linux fundamentals helps beginners develop the technical skills needed for system administration and cybersecurity investigations.

The task provided an overview of the learning objectives and introduced the command-line approach used throughout the room. It established the foundation for learning how to interact with a Linux operating system and perform common operations through terminal commands.

Task 2: Talking to Linux

This task focused on interacting with Linux through the command-line interface (CLI). The CLI allows users to communicate with the operating system by entering commands in a terminal instead of relying entirely on a graphical interface.

Basic commands help users identify their current account, understand their location within the filesystem, and perform simple operations. For example, `whoami` displays the current username, while `pwd` shows the current working directory. The `ls` command lists files and directories, and `cat` displays the contents of a text file.

Learning these commands is important because many cybersecurity tools and administrative tasks require terminal interaction. Command-line skills also help analysts work efficiently when examining files, reviewing system information, and troubleshooting problems.

Task 3: Finding Your Way Around

This task introduced Linux filesystem navigation and explained how to move between directories and view file contents. Linux organizes files and directories in a hierarchical structure, beginning at the root directory.

The `ls` command is used to list the contents of the current directory. The `cd` command changes the working directory, while `pwd` displays the current location. The `cat` command is used to read and display the contents of a file.

These commands help users explore directories, locate relevant files, and understand how information is organized within a Linux system. Filesystem navigation is an essential skill for cybersecurity professionals because investigating systems often involves locating configuration files, application data, and security logs.

Task 4: Let the Machine Do the Searching

This task introduced searching techniques that help users locate files and identify specific information within text. Searching commands are particularly useful when working with large directories or reviewing system logs.

The `find` command searches for files and directories that match specified criteria, such as a filename or location. The `grep` command searches text for a particular word or pattern and displays matching lines.

For example, the command `grep "THM" access.log` searches a file named `access.log` for lines containing the text `THM`. Similarly, searching log files for words such as `Failed` can help an analyst locate entries that may relate to unsuccessful authentication attempts.

These techniques are useful in cybersecurity because analysts often need to examine large amounts of data efficiently. Searching tools can help narrow down relevant information during log analysis, troubleshooting, and initial security investigations. However, search results must be reviewed in context before conclusions are made.

Task 5: Shell Operators (Combining Commands)

This task explained how shell operators can be used to control command execution and redirect command output. These operators make it possible to combine operations and manage information directly through the terminal.

The `&` operator can run a command in the background, allowing the shell to accept further input. The `&&` operator executes the next command only when the preceding command succeeds. These operators help users control how commands are executed.

The `>` operator redirects output into a file and replaces its existing contents. In contrast, the `>>` operator appends output to the end of a file without overwriting the existing content.

For example, `echo "TryHackMe" > thm` writes text to a file named `thm`, replacing any existing contents. The command `echo "thm" >> thm` adds more text to the same file. The `cat thm` command can then be used to display the file contents.

Understanding shell operators is useful for automating repetitive tasks, saving command results, and combining commands into more efficient workflows. These skills can also support basic scripting and cybersecurity analysis.

Conclusion

The TryHackMe Linux Fundamentals (Pt1) room provided an introduction to essential Linux concepts and command-line operations. Across the five tasks, the learning material covered Linux fundamentals, terminal interaction, filesystem navigation, searching techniques, and shell operators.

The commands `ls`, `cd`, `pwd`, and `cat` support everyday filesystem navigation and file viewing. The `find` and `grep` commands help locate files and search text, while shell operators such as `&`, `&&`, `>`, and `>>` provide greater control over command execution and output handling.

These concepts are relevant to cybersecurity because Linux systems are widely used in servers, security tools, and SOC environments. The ability to navigate directories, search files, and understand log data forms a foundation for future learning in Linux administration, security monitoring, and incident investigation.

Overall, this learning activity established a foundation for developing practical Linux skills and understanding how command-line tools can support cybersecurity tasks.